ARUSH JASUJA

New York, NY • (646) 208-7412 • arushjasuja@gmail.com • linkedin.com/in/arushjasuja • github.com/arushjasuja

Software engineer with 3 years building and debugging ML infrastructure in production: model deployment, model evaluation, optimization, and data processing for GPU-accelerated deep learning systems serving paying customers. MS Computer Science, NYU. 5 peer-reviewed ML publications.

EDUCATION

New York University — MS, Computer Science | GPA: 3.83/4.0 | Sep 2024 – May 2026

Coursework: Graphics Processing Units (Prof. Mohamed Zahran), Deep Learning (Prof. Yann LeCun), Algorithms, Operating Systems

Guru Gobind Singh Indraprastha University — B.Tech, Computer Science | CGPA: 8.0/10 | Sep 2016 – Sep 2020

EXPERIENCE

Lead Software Engineer | Ripik.AI — New Delhi, India | Aug 2022 – Jul 2024

- Cut a customer-facing ML planning pipeline from 14 hours to 2 hours for 8 manufacturing facilities. Profiled CNN/ViT inference with Nsight Compute, fixed uncoalesced memory access (AoS to SoA layout), and moved inference to mixed precision (FP16/BF16); GPU throughput improved roughly 40% on the same hardware, deferring a $15M+ hardware purchase.
- Built multi-node distributed training infrastructure in PyTorch: started on DDP with NCCL, then migrated to FSDP (ZeRO-3) with forward prefetch after tracing AllReduce to 58% of step time. Per-GPU model state dropped 96% (36.8 GB to 1.53 GB across 24 cloud GPUs) and step time fell 27.7%.
- Owned ML infrastructure end to end: model deployment through CI/CD with an 8-minute rollback path, per-class model evaluation gates, and data processing pipelines for sensor and image data. A serving-layer interface I designed cut new-model integration from 3 days to under 4 hours.
- Debugged a silent INT8 quantization regression: rare defect classes lost 12% accuracy, invisible at 0.24% overall. Root-caused calibration range clipping, shipped stratified calibration plus per-class accuracy gates that blocked the failure mode permanently.
- Mentored junior engineers on profiling methodology; one engineer's first solo optimization raised a kernel's memory throughput 3.2x against a CI-tracked baseline.

Software Engineer, AI/ML | Sequoia Insilico — New Delhi, India | May 2021 – Aug 2022

- Reduced CNN and LSTM inference latency 45% across 1M+ patient records using PyTorch profiling and INT8 post-training quantization, holding COVID-19 detection accuracy at the 90% clinical floor. Oversampled the rare positive class during calibration to keep per-class accuracy from silently degrading.

PROJECTS

GPU-Optimized N-Body Simulation (NYU, Prof. Zahran's GPU course) | Dec 2025

- 6,095x speedup for a 100K-particle simulation on an RTX 4070. Ran roofline analysis before writing any GPU code (1.67 FLOP/byte vs 58 FLOP/byte ridge: memory-bound by 35x); the AoS-to-SoA layout change alone measured 4.26x against a ~3x analytical prediction, with the gap explained by TLB fragmentation. Final kernels hit 60% compute / 52% memory utilization in Nsight Compute.

Decoder-Only Transformer From Scratch — PyTorch and JAX (NYU Deep Learning) | 2025

- Implemented and trained a GPT-style decoder in PyTorch and JAX: multi-head attention, RoPE, KV-cache inference, BPE tokenizer, and a mixed-precision multi-GPU data-parallel training loop. Matched reference perplexity on the course corpus and wrote unit tests for attention masking and cache correctness.

PUBLICATIONS (5 peer-reviewed ML publications - https://scholar.google.com/citations?user=Sc1Mkb8AAAAJ)

"Connecting Dots with Deep Learning: Graph-Based Approach of Alzheimer's Conversion Prediction" | Alzheimer's & Dementia 2025 | Impact Factor: 11.1

"Layer-wise Adaptive Sine Activation Based Recurrent Network for MCI Conversion" | Alzheimer's & Dementia 2025 | Impact Factor: 11.1

TECHNICAL SKILLS

Languages: Python, C++ (CUDA C/C++), C, SQL, Bash
ML frameworks: PyTorch, JAX, TensorFlow

ML infrastructure: model deployment, model evaluation, optimization, data processing, debugging, distributed training (DDP, FSDP/ZeRO, NCCL), mixed precision (FP16/BF16), quantization (INT8, FP8), TensorRT, vLLM, Triton kernels, CI/CD, Docker, Kubernetes

Systems: large-scale distributed systems, GPU architecture, profiling (Nsight Compute/Systems), roofline analysis, memory optimization, data structures and algorithms, Linux
Leadership: Mentored 15+ engineers; speaker at 3 conferences; NYU course section leader (Architecture, Algorithms)